Justin Payan Curriculum Vitae

Carnegie Mellon University
Machine Learning Department
Pittsburgh, PA

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Employment

10/2024 – present **Carnegie Mellon University**
Postdoctoral Research Associate with Nihar B. Shah
Pittsburgh, PA
Current projects include grecaptcha.com and auditing AI science workflows for vulnerability to backdoor attacks.

Education

09/2018 – 02/2025 **University of Massachusetts Amherst**
Ph.D. in Computer Science
Advisor: Yair Zick
Thesis: Data Driven Expert Assignment
Committee: Nihar B. Shah, Andrew McCallum, Andrew McGregor

09/2013 – 05/2018 **University of Georgia**
M.S. in Artificial Intelligence, B.S. in Mathematics, A.B. in Cognitive Science
Advisor: Frederick Maier

Past Employment

05/2023 – 08/2023 **Adobe Research**, Document Intelligence Lab
Research Intern with Chris Tensmeyer

06/2022 – 09/2022 **Microsoft**, Calc Intelligence
Data Science Intern with Kartik Sridhar

06/2021 – 09/2021 **Amazon Alexa**, Trustworthy Alexa
Research Intern with Yuval Merhav

05/2020 – 08/2020 **Amazon Alexa**, Trustworthy Alexa
Research Intern with Yuval Merhav

06/2017 – 06/2018 **Vertica**, Machine Learning
Software Engineer with Vincent Xu

Awards, Grants, and Scholarships

2026 – 2027 **Alfred P. Sloan Foundation**
Co-PI, "Reliable Autonomous AI Scientists."
With Nihar B. Shah and Atoosa Kasirzadeh.
Amount: $300,000

Spring 2026 **Google Cloud Computing Credits**
Equivalent of $30,000.

2013 – 2017 **University of Georgia Foundation Fellowship**
$128,260 in tuition and stipend for housing, research, and travel.

Peer-Reviewed Publications

Google Scholar: https://scholar.google.com/citations?user=gc9w_eYAAAAJ&hl=en
* denotes equal contribution; $(\alpha - \beta)$ denotes alphabetical author order.

Journal Publications

[J2] Sarina Xi, Vishisht Rao, **Justin Payan**, and Nihar B. Shah. FLAWS: A Benchmark for Error Identification and Localization in Scientific Papers. *DMLR 2026 (Forthcoming)*. [link]

[J1] Hadi Hosseini, Andrew McGregor, **Justin Payan**, Rik Sengupta, Rohit Vaish, and Vignesh Viswanathan $(\alpha - \beta)$. Graphical House Allocation with Identical Valuations. *JAAMAS 2024*. [link]

Conference Publications

[C8] **Justin Payan**, Noemi Barbagli, Ignacio Toledo, Rodrigo A. Carrasco, Sergio Martín, and Nihar B. Shah. Uncertainty-aware Ground-based Telescope Observation Scheduling at ALMA. *SPIE Astronomical Telescopes and Instrumentation 2026*. [link]

[C7] Elita Lobo*, **Justin Payan***, Cyrus Cousins, and Yair Zick. Fair and Welfare-Efficient Constrained Multi-matchings under Uncertainty. *NeurIPS 2024*. [link]

[C6] **Justin Payan**, Swaroop Mishra, Mukul Singh, Carina Negreanu, Christian Poelitz, Chitta Baral, Subhro Roy, Rasika Chakravarthy, Benjamin Van Durme, and Elnaz Nouri. InstructExcel: A Benchmark for Natural Language Instruction in Excel. *EMNLP 2023 (Findings)*. [link]

[C5] Cyrus Cousins, **Justin Payan**, and Yair Zick $(\alpha - \beta)$. Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities. *SAGT 2023*. [link]

[C4] **Justin Payan**, Rik Sengupta, and Vignesh Viswanathan $(\alpha - \beta)$. Relaxations of Envy-Freeness over Graphs. *AAMAS 2023 (Extended Abstract)*. [link]

[C3] **Justin Payan** and Yair Zick. I Will Have Order! Optimizing Orders for Fair Reviewer Assignment. *IJCAI 2022*. [link]

[C2] **Justin Payan**, Yuval Merhav, He Xie, Satyapriya Krishna, Anil Ramakrishna, Mukund Sridhar, and Rahul Gupta. Towards Realistic Single-Task Continuous Learning Research for NER. *EMNLP 2021 (Findings)*. [link]

[C1] Ananya Gupta*, Eric Johnson*, **Justin Payan**, Aditya Roy, Ari Kobren, Swetasudha Panda, Michael Wick, and Jean-Baptiste Tristan. Online Post-Processing in Rankings for Fair Utility Maximization. *WSDM 2021*. [link]

Preprints

* denotes equal contribution.

[P2] **Justin Payan***, Bálint Gyevnár*, Atoosa Kasirzadeh, and Nihar B. Shah. greCAPTCHA: Assessing Understanding as Evidence of Research Authorship Under Generative AI. *Under Submission*. [link]

[P1] Vishisht Rao, **Justin Payan**, Andrew McCallum, and Nihar B. Shah. ML Researchers Support Openness in Peer Review But Are Concerned About Resubmission Bias. *Under Submission*. [link]

Workshop Publications

[W1] **Justin Payan**, Nicholas Monath, and Andrew McCallum. Document Representations using Fine-Grained Topics. *Sets & Partitions Workshop at NeurIPS 2019*. [link]

Code Contributions and Deployment

2026 **greCAPTCHA (grecaptcha.com)**
Public-facing platform for assessing understanding of research papers. Generates manuscript-specific questions, evaluates responses, and provides reports for human review.

2026 **PULSAR**
Designed a telescope scheduling algorithm that improves astronomical objective completion while improving satisfaction of global scheduling constraints. Currently being evaluated for use at the ALMA Telescope.

2022 **OpenReview: FairSequence**
Created and deployed the FairSequence reviewer matching algorithm on the OpenReview platform, used by over 50 peer review venues.

Teaching

Instructor of Record

Fall 2020, Fall 2021 **CMPSCI 590N/CICS 580 Intro to Numerical Computing with Python**
University of Massachusetts Amherst

Graduate Teaching Assistant

Spring 2021 **CMPSCI 611 Advanced Algorithms**
University of Massachusetts Amherst

Fall 2020 **CMPSCI 121 Intro to Problem Solving with Computers**
University of Massachusetts Amherst

Fall 2018 **CMPSCI 383 Intro to AI**
University of Massachusetts Amherst

Mentoring

Undergraduate Students

Summer '21 **Rudra Gopal**
University of Massachusetts. Now: MS at Georgia Tech.

Summer '21 **Gautam Chatterjee**
University of Massachusetts. Now: Software Engineer at Wayfair.

Summer '21 **Ed Almusalamy**
University of Massachusetts. Now: MS at University of Massachusetts.

2023-2024 **Hassan Gudal**
University of Massachusetts. Graduated 2024.

2025-2026 **Noemi Barbagli**
Carnegie Mellon University.

2026-present **Yiming Shao**
Carnegie Mellon University.

Master's Students

2020-2021 **Eric Johnson**
University of Massachusetts. Graduated 2021.

2020-2021 **Ananya Gupta**
University of Massachusetts. Now: MLE at Diffbot.

2020-2021 **Aditya Kumar Roy**
University of Massachusetts. Graduated 2021.

2022-2023 **Vedant Bhatia**
University of Massachusetts. Now: Senior Software Engineer at ElisaAI.

2024-2026 **Vishisht Rao**
Carnegie Mellon University. Now: PhD student at Carnegie Mellon University.

2024-2025 **Sarina Xi**
Carnegie Mellon University. Now: PhD student at University of Wisconsin-Madison.

2024-2025 **Arshika Lalan**
Carnegie Mellon University. Now: Senior ML Scientist at JPMC.

2026-present **Kerissa Duliga**
Carnegie Mellon University.

Service

2023 **Workflow Chair**
IJCAI 2023

2023 – 2025 **Workshop Co-organizer**
Computational Fair Division, IJCAI 2023–2025

2025 **Workshop Co-organizer**
Practical Improvements to Peer Review, AAMAS 2025
Workshop on Incentives in Academia, EC 2025

2023, 2025 **Program Committee / Reviewer**
ACL 2023; EC 2025; KDD 2025; CHI 2027

2022, 2024 **Subreviewer**
NeurIPS 2024; EC 2024; AAMAS 2022; GAIW at AAMAS 2022

Media Coverage

09/2026 **Times Higher Education**. Frances Jones.
Researchers submitting AI-assisted papers they 'don't understand'.
September 17, 2026. Coverage of greCAPTCHA.

Invited Talks

10/2026 **University of Pittsburgh**, NLP Seminar
Hosted by Xiang Lorraine Li
AI in Science: Foundations for an Explosion of Discovery

10/2026 **Rutgers University**
Hosted by Arpita Biswas
AI in Science: Foundations for an Explosion of Discovery

10/2026 **University of Washington**, Data Science and AI Seminar
Hosted by eScience Institute
AI in Science: Foundations for an Explosion of Discovery

03/2023 **Brown University**, E-GLAMOR Group
Hosted by Amy Greenwald
Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities

01/2023 **Harvard University**, EconCS Seminar
Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities

11/2022 **Carnegie Mellon University**, Nihar Shah's Group
Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities

04/2021 **UMass Data Analytics and Computational Social Science Brownbag Series**
Envy-Freeness in Paper Reviewer Assignment

04/2021 **UMass Amherst Theory Seminar**
Fair Reviewer Assignment